

$$2.C) R(\rho) = \sum_{a \in \{I, X, Z, XZ\}} U_a^{-1} P_a \rho P_a U_a + P_{\perp} \rho P_{\perp}$$

$$U_a \in \{I, X, Z, XZ\}$$

P = Projector onto the codespace.

$$= |0_L\rangle\langle 0_L| + |1_L\rangle\langle 1_L|$$

So,

U_a	P_a
I	$I P I^{\dagger}$
X	$X P X^{\dagger}$
Z	$Z P Z^{\dagger}$
XZ	$X Z P Z^{\dagger} X^{\dagger}$

The projectors in the recovery channel measure which syndrome sector 'a' is occupied and U_a^{-1} rotates that sector back to the codespace.

The term $P_{\perp} \rho P_{\perp}$ is just to preserve parts of the input state that are not reachable by these correctable errors.

Experimentally:

- 1) We will first measure the syndrome.
- 2) Identify the corresponding error sector from the syndrome.
- 3) Apply a sequence of corrective gates to the errored states since applying the same unitary twice will give you I .

For this code the correction is

Ua	Ua^{-1}
I	I
X	$X^{-1} = X^7$
Z	$Z^{-1} = Z^7$
XZ	$(XZ)^{-1} = X^7 \cdot Z^7$

... QED.